Experiment 23: Hypothesis Testing for a Medical Treatment

Aim: To compare treatment and control groups using a t-test, calculate the p-value, and visualize the results.

Algorithm:

1. Import NumPy, Matplotlib, and the t-test function.
2. Store control and treatment group data.
3. Perform an independent t-test.
4. Calculate and display the p-value.
5. Plot a bar graph comparing the group means.
6. If p-value < 0.05, display a significant treatment effect; otherwise, display no significant effect.

Input:

control = [5, 6, 7, 5, 6]

treatment = [8, 9, 7, 10, 9]

Code:

Experiment 23 – Hypothesis Testing

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind

control = np.array([5, 6, 7, 5, 6])
treatment = np.array([8, 9, 7, 10, 9])

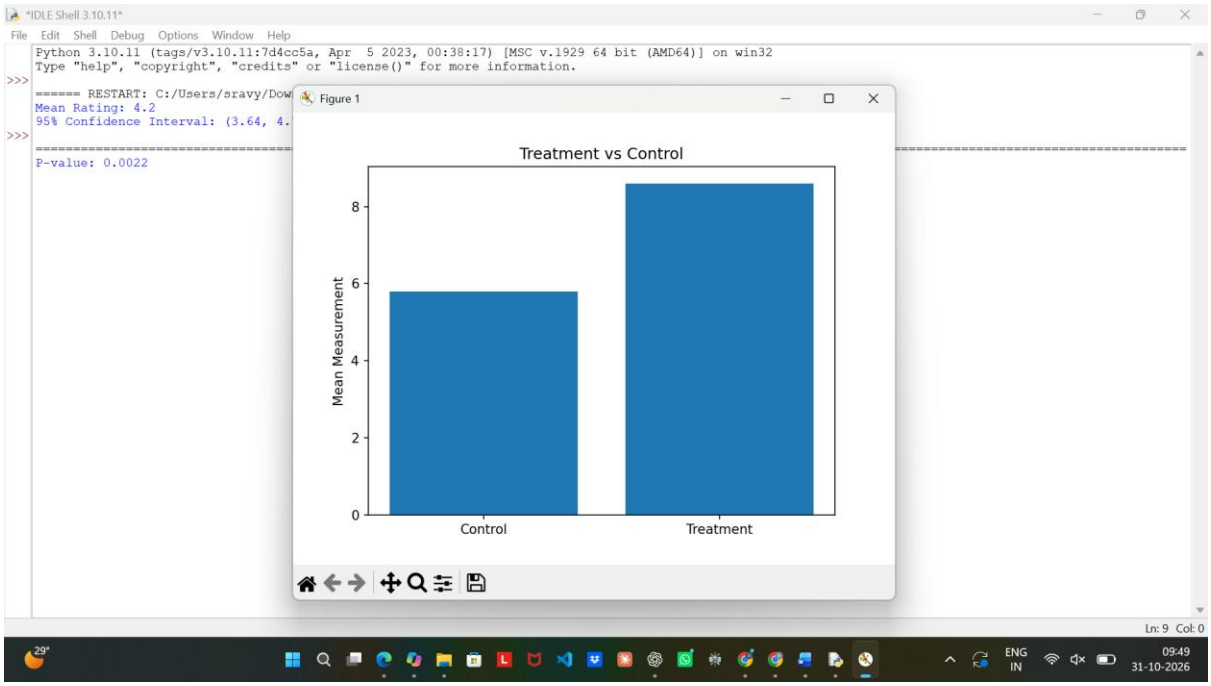
t, p = ttest_ind(control, treatment)

print("P-value:", round(p, 4))

plt.bar(["Control", "Treatment"],
        [control.mean(), treatment.mean()])
plt.ylabel("Mean Measurement")
plt.title("Treatment vs Control")
```

```
plt.show()


if p < 0.05:
    print("Significant treatment effect")
else:
    print("No significant treatment effect")
```



Result: The t-test p-value was calculated, and a bar graph was generated to compare the two groups.

Note: These examples use illustrative data. Use the actual experimental measurements for valid conclusions.